(Please write your Roll Number) Roll No. .....

First-Term Examination

Sixth Semester (B.Tech) April-2022

Paper Code: ETCS-302 Subject: Compiler Design

Time: 1 Hr 30 Mins Max. Marks: 30

Note: Q1 is compulsory and attempt any two questions out of remaining three.

Q1 Attempt all five parts (5*2 =10)

a) What is a cross compiler and how it is different from native compiler?

b) What is the need for separating the parser from scanner?

c) Find leftmost derivation, rightmost derivation and derivation tree of a string 'aacbbcc' from following grammar productions {S→aAcB|Bds , B→aAcA|cAB|b , A→aB|aBc|a}.

d) What are the responsibilities of Loader, Linker and Assembler in the compiler environment?

e) Write steps involved in compilation of following sentence

pos = interest + rate * 60

Q2 Attempt both parts (2*5=10)

a) Remove indirect left recursion from the following grammar:

S → Aa | b

A→Ac | Sd | e

b) Prove that the following grammar is ambiguous and convert it to unambiguous one.

S-> S+S|S*S|(S)|a

Q 3 (1*10=10)

Write Algorithm to construct predictive parser. Design a predictive parser for the given grammar and show the processing of one string.

S→aABb

A→c|∈

B→d|∈

Q 4 Attempt both parts (2*5=10)

a) Draw the finite automata which can be used by lexical analyzer to recognize identifier, number and operators.

b) What is operator grammar? Write operator precedence parsing table for following grammar:

S→(L)|a

L→L ,S|S
